ITRACONAZOLE

Heart failure warning

Strong CYP3A4 inhibitor

CHEMICAL NAME (±)-cis-4-[4-[4-[4-[[2-(2,4-dichlorophenyl)-2-(1H-1,2,4-triazol-1-ylmethyl)-1,3-dioxolan-4-yl]methoxy]phenyl]piperazin-1-yl]phenyl]-2-(butan-2-yl)-2,4-dihydro-3H-1,2,4-triazol-3-one

CLASS Triazole antifungal (broad spectrum)

TRADE NAMES Sporanox · Itrasys · Itrazol

THERAPEUTIC USES / INDICATIONS

- Onychomycosis (pulse or continuous therapy) and tinea infections.
- Pityriasis versicolor; vulvovaginal and oral candidiasis.
- Aspergillosis (chronic, ABPA), histoplasmosis, blastomycosis, sporotrichosis.
- Antifungal prophylaxis in haematological malignancy.

ADMINISTRATION

- **Capsules:** with a full meal (needs acid); avoid antacids/PPIs (or take with cola).
- **Oral solution:** on an empty stomach; swish and swallow.
- Capsules and solution are not interchangeable.

DOSAGES

- **Onychomycosis (pulse):** 200 mg twice daily for 1 week each month (2 pulses fingernails, 3 toenails).
- **Tinea corporis/cruris:** 100–200 mg daily for 7–15 days.
- **Vaginal candidiasis:** 200 mg twice daily for 1 day.
- **Aspergillosis:** 200 mg twice daily.

PHARMACOKINETICS

- **Peak:** 2–5 h. Variable absorption.
- Highly lipophilic — concentrates in skin, nails (persists months).
- Metabolised in liver by CYP3A4 to active hydroxy-itraconazole; excreted in faeces. **t½:** 20–60 h.

PHARMACODYNAMICS

- Inhibits fungal **14α-demethylase** → ↓ ergosterol → fungistatic.
- Broad spectrum incl. dermatophytes, *Candida*, *Aspergillus*, dimorphic fungi.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Nausea, abdominal pain, dyspepsia.
- Headache, rash.
- Hypokalaemia, oedema.

ADVERSE EFFECTS (SERIOUS)

- Congestive heart failure (negative inotrope).
- Hepatotoxicity (rarely fatal).
- Peripheral neuropathy.
- QT prolongation.

CONTRAINDICATIONS & CAUTIONS

- Ventricular dysfunction / heart failure (for nail infections).
- Use with many CYP3A4 substrates: simvastatin, lovastatin, midazolam (oral), pimozide, quinidine, ergot alkaloids, colchicine (in renal/hepatic impairment).
- Pregnancy (for non-life-threatening infections).

KEY INTERACTIONS

- **CYP3A4 substrates:** ↑ levels (statins → rhabdomyolysis; calcium-channel blockers → oedema, heart failure).
- **PPIs, H2 blockers, antacids:** ↓ capsule absorption.
- **Rifampicin, phenytoin:** ↓ itraconazole.

NURSING MANAGEMENT

- Screen for heart failure symptoms and interacting drugs.
- Give capsules with food; solution on empty stomach.
- Monitor LFTs (long courses), weight, oedema, K^+.
- Assess nail/skin response (takes months).
- Advise contraception.

PATIENT EDUCATION

- Take capsules right after a full meal.
- Report swelling of legs, breathlessness, yellow skin, dark urine.
- Nails take months to look normal; continue pulses as prescribed.
- Use contraception during and 2 months after.

KEY POINTS

- Triazole for nails, skin, aspergillosis.
- Avoid in heart failure.
- Capsules with food; solution empty stomach.
- Strong CYP3A4 inhibitor.